

Table 5.6 MEMOBUS/Modbus Communications Command Data

Register No. (Hex.)	Description	
0000	Reserved	
0001	Run command, multi-function input command	
	bit 0	When $H5-12 = 0$, Forward run/stop 1: Forward run, 0: Stop When $H5-12 = 1$, run/stop 1: Run, 0: Stop
	bit 1	When $H5-12 = 0$, Reverse run/stop 1: Reverse run, 0: Stop When $H5-12 = 1$, Forward/Reverse run 1: Reverse, 0: Forward run
	bit 2	External fault 1: EF0 [Option Card External Fault]
	bit 3	Fault Reset 1: Reset command
	bit 4	Multi-function input command 1 When $H1-01 = 40$ [Forward Run Command (2-Wire Seq)], the multi-function input command is "ComRef." Note: When the bit is switched ON as ComRef, the frequency reference source changes to MEMOBUS/Modbus communications. However, the frequency reference source gives priority to the communications option when the communication option is connected to the drive.
	bit 5	Multi-function input command 2 When the multi-function input command is $H1-02 = 41$ [Reverse Run Command (2-Wire Seq)], bit 5 is "ComCtrl." Note: When the bit is switched ON as ComCtrl, the Run command source changes to MEMOBUS/Modbus communications. However, the Run command source gives priority to the communications option when the communication option is connected to the drive.
	bit 6	Multi-function input command 3
	bit 7	Multi-function input command 4
	bit 8	Multi-function input command 5
	bit 9	Multi-function input command 6
	bit A	Multi-function input command 7
	bit B	Multi-function input command 8
	bit C - F	Reserved
0002	Frequency reference	The units are determined by $o1-03$ [Keypad Display Selection] (unsigned).
0003	Output voltage gain	Units: 0.1% Setting range: 20 (2.0%) to 2000 (200.0%), the default value at power up: 1000 (100.0%)
0004	Torque reference/torque limit (0.1% signed)	
0005	Torque compensation (0.1% signed)	
0006	PID setpoint (0.01% signed)	
0007	Setting for the multi-function analog monitor output terminal 1 (10 V/4000 H)	
0008	Setting for the multi-function analog monitor output terminal 2 (10 V/4000 H)	

5.3 MEMOBUS/Modbus Communications

Register No. (Hex.)	Description	
0009	MFDO setting	
	bit 0	MFDO (terminal M1-M2) 1: ON, 0: OFF
	bit 1	MFDO (terminal M3-M4) 1: ON, 0: OFF
	bit 2	MFDO (terminal M5-M6) 1: ON, 0: OFF
	bit 3 - 5	Reserved
	bit 6	1: bit 7 function is enabled
	bit 7	Fault relay output (terminal MA/MB-MC) 1: ON, 0: OFF
	bit 8 - F	Reserved
000A	Pulse train output (Units: 1/1 Hz, setting range: 0 to 32000)	
000B - 000E	Reserved	
000F	Command selection setting	
	bit 0	Reserved
	bit 1	Input for the PID setpoint 1: Target values from the MEMOBUS/Modbus are enabled
	bit 2	Torque reference/torque limit input 1: Setting values from the MEMOBUS/Modbus are enabled
	bit 3	Torque Compensation Input 1: Setting values from the MEMOBUS/Modbus are enabled
	bit 4	Reserved
	bit 5	PID feedback from the MEMOBUS/Modbus 1: PID feedback (15FF (Hex.)) from the MEMOBUS/Modbus is enabled
	bit 6 - B	Reserved
	bit C	Terminal S5 input of broadcast message 1: Enabled, 0: Disabled
	bit D	Terminal S6 input of broadcast message 1: Enabled, 0: Disabled
	bit E	Terminal S7 input of broadcast message 1: Enabled, 0: Disabled
	bit F	Terminal S8 input of broadcast message 1: Enabled, 0: Disabled
0010 - 001A	Reserved	
001B	Analog monitor option AO-A3 analog output 1 value (10 V/4000 (Hex.))	
001C	Analog monitor option AO-A3 analog output 2 value (10 V/4000 (Hex.))	
001D	Digital output option DO-A3 output value (binary)	
001E - 001F	Reserved	
15C0	bit 0	Extended multi-function input command 1
	bit 1	Extended multi-function input command 2
	bit 2	Extended multi-function input command 3
	bit 3 - F	Reserved

■ Monitor Data

Monitor data can only be read.

Table 5.7 Monitor Data for MEMOBUS/Modbus Communication

Register No. (Hex.)	Description	
0020	Drive status 1	
	bit 0	During Run: 1: During run, 0: During stop
	bit 1	During Reverse 1: During reverse, 0: Forward run
	bit 2	Drive ready 1: Ready, 0: Not ready
	bit 3	Fault 1: Fault
	bit 4	Data Setting Error 1: oPExx error
	bit 5	MFDO (terminal M1-M2) 1: ON, 0: OFF
	bit 6	MFDO (terminal M3-M4) 1: ON, 0: OFF
	bit 7	MFDO (terminal M5-M6) 1: ON, 0: OFF
	bit 8 - D	Reserved
	bit E	ComRef status 1: Enabled
	bit F	ComCtrl status 1: Enabled
0021	Fault description 1	
	bit 0	oC [Overcurrent], GF [Ground Fault]
	bit 1	ov [DC Bus Overvoltage]
	bit 2	oL2 [Drive Overloaded]
	bit 3	oH1 [Heatsink Overheat], oH2 [Drive Overheat Warning]
	bit 4	rH [Braking Resistor Overheat], tr [Dynamic Braking Transistor]
	bit 5	Reserved
	bit 6	FbL [PID Feedback Loss], FbH [Excessive PID Feedback]
	bit 7	EF0 [Option Card External Fault], EF1 to EF8 [External fault]
	bit 8	CPFxx [Hardware Fault] Note: Includes oFx.
	bit 9	oL1 [Motor Overload], oL3, L4 [Overtorque Detection 1/2], UL3, L4 [Undertorque Detection 1/2]
	bit A	PGo [PG Disconnect], PGoH [PG Hardware Fault], oS [Overspeed], dEv [Speed Deviation]
	bit B	During Uv [DC Bus Undervoltage] detection
	bit C	Uv1 [DC Bus Undervoltage], Uv2 [Ctrl Power Supply Voltage Fault], Uv3 [SoftCharge Bypass Circuit Fault]
	bit D	LF [Output Phase Loss], PF [Input Phase Loss]
	bit E	CE [MEMOBUS/Modbus Communications Error], bUS [Option Communication Error]
	bit F	oPr [Keypad Connection Fault]

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Register No. (Hex.)	Description	
0022	Fault Contents	
	bit 0	1: During data writing, during motor switching
	bit 1	Reserved
	bit 2	
	bit 3	1: Upper/Lower Limit Fault
	bit 4	1: Data Integrity Fault
	bit 5	1: During EEPROM writing
	bit 6	0: EEPROM writing 1: Change only data on the RAM Note: Enabled when <i>H5-17 = 1</i> [<i>Busy Enter Selection = Write in RAM only</i>].
	bit 7 - F	Reserved
0023	U1-01 [Freq Reference] Note: The unit changes depending on the setting of <i>o1-03</i> [<i>Keypad Display Selection</i>].	
0024	U1-02 [Output Frequency] Note: The unit changes depending on the setting of <i>o1-03</i> [<i>Keypad Display Selection</i>].	
0025	U1-06 [OutVoltage Reference] (units: 0.1 V) Note: Able to replace the setting unit with <i>H5-10</i> [<i>Unit Sel for MEMOBUS/Modbus 0025H</i>].	
0026	U1-03 [Output Current] Note: The unit of display varies depending on the model. 2004 to 2042, 4002 to 4023: 0.01 A 2056 to 2415, 4031 to 4675: 0.1 A	
0027	U1-08 [Output Power]	
0028	U1-09 [Torque Reference]	
0029	Fault description 2	
	bit 0	Reserved
	bit 1	GF [Ground Fault]
	bit 2	PF [Input Phase Loss]
	bit 3	LF [Output Phase Loss]
	bit 4	rH [Braking Resistor Overheat]
	bit 5	Reserved
	bit 6	oH4 [Motor Overheat Fault (PTC Input)]
	bit 7 - F	Reserved

Register No. (Hex.)	Description	
002A	Minor Fault Description 1	
	bit 0 - 1	Reserved
	bit 2	EF [FWD/REV Run Command Input Error]
	bit 3	bb [Baseblock]
	bit 4	oL3 [Overtorque 1]
	bit 5	oH [Heatsink Overheat]
	bit 6	ov [DC Bus Overvoltage]
	bit 7	Uv [DC Bus Undervoltage]
	bit 8	FAn [Internal Fan Fault]
	bit 9	CE [MEMOBUS Communication Error]
	bit A	bUS [Option Communication Error]
	bit B	UL3/UL4 [Undertorque Detection 1/2]
	bit C	oH3 [Motor Overheat (PTC)]
	bit D	FbL [PID Feedback Loss], FbH [Excessive PID Feedback]
	bit E	Reserved
	bit F	CALL [Serial Comm Transmission Error]
002B	U1-10 [Input Terminal Status]	
	bit 0	1: Control circuit terminal S1 ON
	bit 1	1: Control circuit terminal S2 ON
	bit 2	1: Control circuit terminal S3 ON
	bit 3	1: Control circuit terminal S4 ON
	bit 4	1: Control circuit terminal S5 ON
	bit 5	1: Control circuit terminal S6 ON
	bit 6	1: Control circuit terminal S7 ON
	bit 7	1: Control circuit terminal S8 ON
	bit 8 - F	Reserved

5.3 MEMOBUS/Modbus Communications

Register No. (Hex.)	Description	
002C	Drive status 2	
	bit 0	During Run: 1: During run
	bit 1	During zero speed 1: During zero speed
	bit 2	Speed agreement 1: During agreement
	bit 3	User-defined speed agreement 1: During agreement
	bit 4	Frequency Detection 1 1: Output frequency $\leq$ L4-01
	bit 5	Frequency Detection 2 1: Output frequency $\geq$ L4-01
	bit 6	Drive ready 1: Run ready
	bit 7	During low voltage detection 1: During detection
	bit 8	During baseblock 1: Drive output during baseblock
	bit 9	Frequency reference mode 1: No communication option, 0: Communication option
	bit A	Run command mode 1: No communication option, 0: Communication option
	bit B	During overtorque/undertorque 1, 2 detection
	bit C	Frequency reference loss 1: Loss
	bit D	Restart Enabled 1: Restart Enabled
	bit E	Fault 1: Fault generated
	bit F	MEMOBUS/Modbus communications timeout 1: At Timeout
002D	U1-11 [Output Terminal Status]	
	bit 0	MFDO (terminal M1-M2) 1: ON, 0: OFF
	bit 1	MFDO (terminal M3-M4) 1: ON, 0: OFF
	bit 2	MFDO (terminal M5-M6) 1: ON, 0: OFF
	bit 3 - 6	Reserved
	bit 7	Fault relay output (terminal MA/MB-MC) 1: ON, 0: OFF
	bit 8 - F	Reserved
002E	Reserved	
002F	Frequency reference bias (Up 2/Down 2 function) (Units: 0.1%)	
0030	Reserved	
0031	U1-07 [DC Bus Voltage] (Units: 1 V)	
0032	U1-09 [Torque Reference] (Units: 1%)	

Register No. (Hex.)	Description	
0033	Reserved	
0034	Product code 1 [ASCII], product type (GA700 =0A)	
0035	Product code 2 [ASCII], region	
0036 - 0037	Reserved	
0038	PID Feedback: Unsigned, input is equivalent to 100%/maximum output frequency (Units:0.1%)	
0039	PID Input: Unsigned, $\pm 100\%$ / $\pm$ maximum output frequency (Units:0.1%)	
003A	PID Output: Unsigned, $\pm 100\%$ / $\pm$ maximum output frequency (Units:0.1%)	
003B - 003C	Reserved	
003D	Communications error description Note: The description of the communications error is saved until the fault is reset.	
	bit 0	CRC Error
	bit 1	Data Length Error
	bit 2	Reserved
	bit 3	Parity Error
	bit 4	Overflow Error
	bit 5	Framing Error
	bit 6	Timeout
	bit 7 - F	Reserved
003E	Output frequency	Units: min^{-1} or r/min Note: Set E2-04, E4-04, E5-04, E9-08 [Number of Motor Poles].
003F		0.01%
0040 - 004A	Used with U1-xx [Status Monitor]. Refer to the U Monitor for parameter details.	
004B	U1-12 [Drive Status]	
	bit 0	1: During run
	bit 1	1: During zero speed
	bit 2	1: During reverse
	bit 3	1: During reset signal input
	bit 4	1: During speed agreement
	bit 5	1: Drive operation ready
	bit 6	1: Minor Fault
	bit 7	1: Fault
	bit 8	1: oPExx [Operation Error] generation
	bit 9	1: Recovery from momentary power loss, 0: Power recovery
	bit A	1: Motor 2 Selection
	bit B	Reserved
	bit E	ComRef status/ NetRef status
	bit F	ComCtrl status/ NetCtrl status
004C - 007E	Use with U1-xx, U4-xx, U5-xx, U6-xx [Monitor]. Refer to "U2: Fault Trace" and "U3: Fault History" for details.	
007F	Minor fault code (Refer to "Minor fault description" for more information on the minor fault codes.)	
0080 - 0097	Use with U2-xx, U3-xx [Monitor]. Refer to "U Monitor," and refer to "Fault Trace/Fault History Descriptions" for details on register values.	
0098 - 0099	U4-01 [Cumulative Ope Time] (Ex.) When U4-01 [Cumulative Ope Time] is 12345, 0098(Hex.) = 1234 and 0099(Hex.) = 5.	

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Register No. (Hex.)	Description	
009A - 009B	U4-03 [Cooling Fan Ope Time] (Ex.) When U4-03 [Cooling Fan Ope Time] is 12345, 009A(Hex.) = 1234 and 009B (Hex.) = 5.	
009C - 00AA	Reserved	
00AB	Drive rated current Note: The unit of display varies depending on the model. 2004 to 2042, 4002 to 4023: 0.01 A 2056 to 2415, 4031 to 4675: 0.1 A	
00AC	U1-05 [Motor Speed]	Units: min ⁻¹ or r/min Note: Set E2-04, E4-04, E5-04, E9-08 [Number of Motor Poles].
00AD		Units: 0.01%
00AE, 00AF	Reserved	
00B0	Optional codes connected to CN5-A	Optional connected codes are stored in the register. AI-A3 = 0003(Hex.) AO-A3 = 0004(Hex.) DI-A3 = 0001(Hex.) DO-A3 = 0002(Hex.) PG-B3 = 0011(Hex.) PG-F3 = 0021(Hex.) PG-RT3 = 0023(Hex.) PG-X3 = 0012(Hex.) SI-C3 = 5343(Hex.) SI-EM3 = 1005(Hex.) SI-EN3 = 1006(Hex.) SI-ET3 = 1004(Hex.) SI-N3 = 534E(Hex.) SI-P3 = 5350(Hex.) SI-S3 = 5353(Hex.) SI-T3 = 5354(Hex.) SI-W3 = 1003(Hex.)
00B1	Reserved	
00B2	Optional codes connected to CN5-B	
00B3	Optional codes connected to CN5-C	
00B4	Reserved	
00B5	U1-16 [Output Freq afterSFS]	Units: min ⁻¹ or r/min Note: Set E2-04, E4-04, E5-04, E9-08 [Number of Motor Poles].
00B6		Units: 0.01%
00B7	Frequency reference monitor	Units: min ⁻¹ or r/min Note: Set E2-04, E4-04, E5-04, E9-08 [Number of Motor Poles].
00B8		Units: 0.01%
00B9 - 00BE	Reserved	
00BF	Operation error number xx of oPExx is displayed.	

Register No. (Hex.)	Description	
00C0	Fault description 3	
	bit 0	Reserved
	bit 1	Uv1 [DC Bus Undervoltage]
	bit 2	Uv2 [Ctrl Power Supply Voltage Fault]
	bit 3	Uv3 [SoftCharge Bypass Circuit Fault]
	bit 4	SC [Out Short Circuit or IGBT Fault]
	bit 5	GF [Ground Fault]
	bit 6	oC [Overcurrent]
	bit 7	ov [DC Bus Overvoltage]
	bit 8	oH [Heatsink Overheat]
	bit 9	oH1 [Heatsink Overheat]
	bit A	oL1 [Motor Overload]
	bit B	oL2 [Drive Overloaded]
	bit C	oL3 [Overtorque Detection 1]
	bit D	oL4 [Overtorque Detection 2]
	bit E	rr [Dynamic Braking Transistor]
	bit F	rH [Braking Resistor Overheat]
00C1	Fault description 4	
	bit 0	EF3 [External Fault (terminal S3)]
	bit 1	EF4 [External Fault (terminal S4)]
	bit 2	EF5 [External Fault (terminal S5)]
	bit 3	EF6 [External Fault (terminal S6)]
	bit 4	EF7 [External Fault (terminal S7)]
	bit 5	EF8 [External Fault (terminal S8)]
	bit 6	FAn [Internal Fan Fault]
	bit 7	oS [Overspeed]
	bit 8	dEv [Speed Deviation]
	bit 9	PGo [PG Disconnect]
	bit A	PF [Input Phase Loss]
	bit B	LF [Output Phase Loss]
	bit C	oH3 [Motor Overheat Alarm (PTC Input)]
	bit D	oPr [Keypad Connection Fault]
	bit E	Err [EEPROM Write Error]
	bit F	oH4 [Motor Overheat Fault (PTC Input)]

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Register No. (Hex.)	Description	
00C2	Fault description 5	
	bit 0	CE [MEMOBUS/Modbus Communication Fault]
	bit 1	bUS [Option Communication Error]
	bit 2 - 3	Reserved
	bit 4	CF [Control Fault]
	bit 5	SvE [Zero Servo Fault]
	bit 6	EF0 [Option Card External Fault]
	bit 7	FbL [PID Feedback Loss]
	bit 8	UL3 [Undertorque Detection 1]
	bit 9	UL4 [Undertorque Detection 2]
	bit A	oL7 [High Slip Braking oL]
	bit B - E	Reserved
	bit F	Hardware Fault (includes <i>oFx</i> fault)
00C3	Fault description 6	
	bit 0	Reserved
	bit 1	dv1 [Z Pulse Fault]
	bit 2	dv2 [Z Pulse Noise Fault Detection]
	bit 3	dv3 [Inversion Detection]
	bit 4	dv4 [Inversion Prevention Detection]
	bit 5	LF2 [Output Current Imbalance]
	bit 6	STPo [Pull-Out Detection]
	bit 7	PGoH [PG Hardware Fault]
	bit 8	E5 [MECHATROLINK Watchdog Timer Err]
	bit 9	Reserved
	bit A	SEr [Too Many Speed Search Restarts]
	bit B - F	Reserved
00C4	Fault description 7	
	bit 0	FbH [Excessive PID Feedback]
	bit 1	EF1 [External Fault (terminal S1)]
	bit 2	EF2 [External Fault (terminal S2)]
	bit 3	oL5 [Mechanical Weakening Detection 1]
	bit 4	UL5 [Mechanical Weakening Detection 2]
	bit 5	CoF [Current Offset Fault]
	bit 6 - 7	Reserved
	bit 8	dWFL [DriveWorksEZ Fault]
	bit 9	dWF1 [EEPROM Memory DWEZ Data Error]
	bit A - C	Reserved
	bit D	rF [Braking Resistor Fault]
	bit E	boL [BrakingTransistor Overload Fault]
	bit F	Reserved

Register No. (Hex.)	Description	
00C5	Fault description 8	
	bit 0	LSo [LSO Fault]
	bit 1	nSE [Node Setup Error]
	bit 2 - 9	Reserved
	bit A	dv7 [Polarity Judge Timeout]
	bit B - D	Reserved
	bit E	LF3 [Output Phase Loss 3]
	bit F	UnbC [Current Imbalance]
00C6	Fault description 9	
	bit 0	Uv4 [Gate Drive Board Power Supply Voltage Low]
	bit 1 - F	Reserved
00C7	Reserved	
00C8	Minor Fault Description 2	
	bit 0	Uv [DC Bus Undervoltage]
	bit 1	ov [DC Bus Overvoltage]
	bit 2	oH [Heatsink Overheat]
	bit 3	oH2 [Drive Overheat Warning]
	bit 4	oL3 [Overtorque 1]
	bit 5	oL4 [Overtorque 2]
	bit 6	EF [FWD/REV Run Command Input Error]
	bit 7	bb [Baseblock]
	bit 8	EF3 [External Fault (terminal S3)]
	bit 9	EF4 [External Fault (terminal S4)]
	bit A	EF5 [External Fault (terminal S5)]
	bit B	EF6 [External Fault (terminal S6)]
	bit C	EF7 [External Fault (terminal S7)]
	bit D	EF8 [External Fault (terminal S8)]
	bit E	FAn [Internal Fan Fault]
	bit F	oS [Overspeed]

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Register No. (Hex.)	Description	
00C9	Minor Fault Description 3	
	bit 0	dEv [Speed Deviation]
	bit 1	PGo [PG Disconnect]
	bit 2	oPr [Keypad Connection Fault]
	bit 3	CE [MEMOBUS Communication Error]
	bit 4	bUS [Option Communication Error]
	bit 5	CALL [Serial Comm Transmission Error]
	bit 6	oL1 [Motor Overload]
	bit 7	oL2 [Drive Overloaded]
	bit 8	Reserved
	bit 9	EF0 [Option Card External Fault]
	bit A	rUn [Motor Switch during Run]
	bit B	Reserved
	bit C	CALL [Serial Comm Transmission Error]
	bit D	UL3 [Undertorque Detection 1]
	bit E	UL4 [Undertorque Detection 2]
	bit F	SE [MEMOBUS/Modbus Comm TestMode Err]
00CA	Minor Fault Description 4	
	bit 0	Reserved
	bit 1	oH3 [Motor Overheat (PTC)]
	bit 2 - 5	Reserved
	bit 6	FbL [PID Feedback Loss]
	bit 7	FbH [Excessive PID Feedback]
	bit 8	Reserved
	bit 9	dnE [Drive Disabled]
	bit A	PGoH [PG Hardware Fault]
	bit B - F	Reserved
00CB	Minor Fault Description 5	
	bit 0	E5 [MECHATROLINK Watchdog Timer Err]
	bit 1	AEr [Station Address Setting Error (CC-Link, CANopen, MECHATROLINK)]
	bit 2	CyC [MECHATROLINK CommCycleSettingErr]
	bit 3	HCA [Current Alarm]
	bit 4	LT-1 [Cooling Fan Maintenance Time]
	bit 5	LT-2 [Capacitor Maintenance Time]
	bit 6 - 7	Reserved
	bit 8	EF1 [External Fault (terminal S1)]
	bit 9	EF2 [External Fault (terminal S2)]
	bit A	SToF [Safe Disable Signal Input]
	bit B	STo [Safe Disable Signal Input]
	bit C	oL5 [Mechanical Weakening Detection 1]
	bit D	UL5 [Mechanical Weakening Detection 2]
	bit E - F	Reserved

Register No. (Hex.)	Description	
00CC	Minor Fault Description 6	
	bit 0	Reserved
	bit 1	TrPC [IGBT Maintenance Time (90%)]
	bit 2	LT-3 [SoftChargeBypassRelay MainteTime]
	bit 3	LT-4 [IGBT Maintenance Time (50%)]
	bit 4	boL [BrakingTransistor Overload Fault]
	bit 5 - 7	Reserved
	bit 8	dWAL [DriveWorksEZ Fault]
	bit 9 - F	Reserved
00CD - 00CF	Reserved	
00D0	CPF description 1	
	bit 0 - 1	Reserved
	bit 2	CPF02 [A/D Conversion Failure]
	bit 3	CPF03 [PWM Motor Failure]
	bit 4 - 5	Reserved
	bit 6	CPF06 [Control Circuit Error]
	bit 7	CPF07 [Terminal Board Connection Error]
	bit 8	CPF08 [EEPROM Serial Communications Error]
	bit 9	Reserved
	bit A	CRF10 [ASIC Verify Fault]
	bit B	CRF11 [RAM Fault]
	bit C	CRF12 [FLASH Memory Fault]
	bit D	CPF13 [Watchdog Circuit Exception]
	bit E	CPF14 [Control Circuit Fault]
	bit F	Reserved
00D1	CPF description 2	
	bit 0	CPF16 [Clock Fault]
	bit 1	CPF17 [Timing Fault]
	bit 2	CPF18 [Control Circuit Fault]
	bit 3	CPF19 [Control Circuit Fault]
	bit 4	CPF20 [Hardware Fault (at power ON)]
	bit 5	CPF21 [Hardware Fault (after communication start up)]
	bit 6	CPF22 [A/D Conversion Failure]
	bit 7	CPF23 [PWM Feedback Fault]
	bit 8	CPF24 [Drive Capacity Signal Fault]
	bit 9	CPF25 [Terminal Board not Connected]
	bit A	CPF26 [ASIC BB Circuit Error]
	bit B	CPF27 [ASIC PWM Setting Register Error]
	bit C	CPF28 [ASIC PWM Pattern Error]
	bit D	CPF29 [ASIC On-delay Error]
	bit E	CPF30 [ASIC BB ON Error]
	bit F	CPF31 [ASIC Code Error]

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Register No. (Hex.)	Description	
00D2	CPF description 3	
	bit 0	CPF32 [ASIC Start-up Error]
	bit 1	CPF33 [Watchdog Circuit Exception Circuit Fault]
	bit 2	CPF34 [ASIC Power, Clock Error]
	bit 3	CPF35 [External A/D Conversion Failure]
	bit 4	CPU36 [ASIC Communications Error (Initial)]
	bit 5	CPU37 [ASIC Communications Error (Online)]
	bit 6	CPU38 [EEPROM Data Fault (I/F board side)]
	bit 7	CPU39 [CPU-ASIC Communications Fault]
	bit 8	CPF40 [Control Circuit Error]
	bit 9	CPF41 [Control Circuit Error]
	bit A	CPF42 [Control Circuit Error]
	bit B	CPF43 [Control Circuit Error]
	bit C	CPF44 [Control Circuit Error]
	bit D	CPF45 [Control Circuit Error]
	bit E - F	Reserved
00D3 - 00D7	Reserved	
00D8	oFA0x Description (CN5-A)	
	bit 0	oFA00 [Not supported]
	bit 1	oFA01 [Option Connection Error]
	bit 2 - 4	Reserved
	bit 5	oFA05 [A/D Conversion Error]
	bit 6	oFA06 [Option Response Error]
	bit 7 - F	Reserved
00D9	oFA1x Description (CN5-A)	
	bit 0	oFA10 [RAM Fault]
	bit 1	oFA11 [Option Operation Mode Fault (SLMOD)]
	bit 2	oFA12 [CRC Error (Drive receive)]
	bit 3	oFA13 [Frame Error (Drive receive)]
	bit 4	oFA14 [Abort Error (Drive receive)]
	bit 5	oFA15 [CRC Error (Option card receive)]
	bit 6	oFA16 [Frame Error (option card receive)]
	bit 7	oFA17 [Abort Error (option card receive)]
	bit 8 - F	Reserved
00DA	Reserved	

Register No. (Hex.)	Description	
00DB	oFA3x Description (CN5-A)	
	bit 0	oFA30 [Comm. ID Error]
	bit 1	oFA31 [Model Code Error]
	bit 2	oFA32 [Checksum Error]
	bit 3	oFA33 [Comm. option timeout waiting for response]
	bit 4	oFA34 [MEMOBUS/Modbus communications timeout]
	bit 5	oFA35 [Drive timeout waiting for response]
	bit 6	oFA36 [CI Check Error]
	bit 7	oFA37 [Drive timeout waiting for response]
	bit 8	oFA38 [Control Command Selection Error]
	bit 9	oFA39 [Drive timeout waiting for response]
	bit A	oFA40 [Control Response Selection 1 Error]
	bit B	oFA41 [Drive timeout waiting for response]
	bit C	oFA42 [Control Response Selection 2 Error]
	bit D	oFA43 [Drive timeout waiting for response]
	bit E - F	Reserved
00DC	oFb0x Description (CN5-B)	
	bit 0	oFb00 [Not supported]
	bit 1	oFb01 [Option Connection Error]
	bit 2	oFb02 [DuplicateOptions]
	bit 3 - 4	Reserved
	bit 5	oFb05 [A/D Conversion Error]
	bit 6	oFb06 [Option Response Error]
	bit 7 - F	Reserved
00DD	oFb1x Description (CN5-B)	
	bit 0	oFb10 [RAM Fault]
	bit 1	oFb11 [Option Operation Mode Fault (SLMOD)]
	bit 2	oFb12 [CRC Error (Drive receive)]
	bit 3	oFb13 [Frame Error (Drive receive)]
	bit 4	oFb14 [Abort Error (Drive receive)]
	bit 5	oFb15 [CRC Error (Option card receive)]
	bit 6	oFb16 [Frame Error (option card receive)]
	bit 7	oFb17 [Abort Error (option card receive)]
	bit 8 - F	Reserved
00DE - 00DF	Reserved	

5.3 MEMOBUS/Modbus Communications

Register No. (Hex.)	Description	
00E0	oFb3x Description (CN5-B)	
	bit 0	oFb30 [Comm. ID Error]
	bit 1	oFb31 [Model Code Error]
	bit 2	oFb32 [Checksum Error]
	bit 3	oFb33 [Comm. option timeout waiting for response]
	bit 4	oFb34 [MEMOBUS/Modbus communications timeout]
	bit 5	oFb35 [Drive timeout waiting for response]
	bit 6	oFb36 [CI Check Error]
	bit 7	oFb37 [Drive timeout waiting for response]
	bit 8	oFb38 [Control Command Selection Error]
	bit 9	oFb39 [Drive timeout waiting for response]
	bit A	oFb40 [Control Response Selection 1 Error]
	bit B	oFb41 [Drive timeout waiting for response]
	bit C	oFb42 [Control Response Selection 2 Error]
	bit D	oFb43 [Drive timeout waiting for response]
	bit E - F	Reserved
00E1	oFC0x Description (CN5-C)	
	bit 0	oFC00 [Not supported]
	bit 1	oFC01 [Connection Error]
	bit 2	oFC02 [DuplicateOptions]
	bit 3 - 4	Reserved
	bit 5	oFC05 [A/D Conversion Error]
	bit 6	oFC06 [Option Response Error]
00E2	oFC1x Description (CN5-C)	
	bit 0	oFC10 [RAM Fault]
	bit 1	oFC11 [Option Operation Mode Fault (SLMOD)]
	bit 2	oFC12 [CRC Error (Drive receive)]
	bit 3	oFC13 [Frame Error (Drive receive)]
	bit 4	oFC14 [Abort Error (Drive receive)]
	bit 5	oFC15 [CRC Error (Option card receive)]
	bit 6	oFC16 [Frame Error (option card receive)]
	bit 7	oFC17 [Abort Error (option card receive)]
	bit 8 - F	Reserved
00E3	Reserved	
00E4	oFC5x Description (CN5-C)	
	bit 0	oFC50 [Encoder Option AD Conversion Error]
	bit 1	oFC51 [Encoder Option Analog Circuit Error]
	bit 2	oFC52 [Encoder Communication Timeout]
	bit 3	oFC53 [Encoder Communication Data Error]
	bit 4	oFC54 [Encoder Error]
	bit 5	oFC55 [Resolver Error]
	bit 6 - F	Reserved

Register No. (Hex.)	Description
00E5 - 00E9	Reserved
00EA	Fault description 11
	bit 0- D Reserved
	bit E SCF [Safety Circuit Fault]
	bit F Reserved
00EB - 00FA	Reserved
00FB	Output current Note: The unit of display varies depending on the model. 2004 to 2042, 4002 to 4023: 0.01 A 2056 to 2415, 4031 to 4675: 0.1 A

■ Broadcast Messages

Broadcast messages are available as read-only.

The undefined bit signal in the broadcast operation signal continues using the local data signal.

Table 5.8 Broadcast Messages for MEMOBUS/Modbus Communication

Register No. (Hex.)	Description
0001	Operation signal
	bit 0 Run command 1: Run, 0: Stop
	bit 1 Reverse run command 1: Reverse, 0: Forward run
	bit 2 - 3 Reserved
	bit 4 External fault 1: EF0 [Option Card External Fault]
	bit 5 Fault Reset 1: Reset command
	bit 6 - B Reserved
	bit C MFDI terminal S5 input
	bit D MFDI terminal S6 input
	bit E MFDI terminal S7 input
	bit F MFDI terminal S8 input
0002	Frequency reference 30000/100%

■ Fault Trace/Fault History Contents

The following table lists the fault codes read using the commands from monitors [*U2-xx*, *U3-xx*].

Table 5.9 Fault Trace/Fault History Contents

Fault code (Hex.)	Name	Fault code (Hex.)	Name
0002	Uv1 [DC Bus Undervoltage]	0008	ov [DC Bus Overvoltage]
0003	Uv2 [Ctrl Power Supply Voltage Fault]	0009	oH [Heatsink Overheat]
0004	Uv3 [SoftCharge Bypass Circuit Fault]	000A	oH1 [Heatsink Overheat]
0005	SC [Out Short Circuit or IGBT Fault]	000B	oL1 [Motor Overload]
0006	GF [Ground Fault]	000C	oL2 [Drive Overloaded]
0007	oC [Overcurrent]	000D	oL3 [Overtorque Detection 1]

5.3 MEMOBUS/Modbus Communications

Fault code (Hex.)	Name
000E	oL4 [Overtorque Detection 2]
000F	rr [Dynamic Braking Transistor]
0010	rH [Braking Resistor Overheat]
0011	EF3 [External Fault (terminal S3)]
0012	EF4 [External Fault (terminal S4)]
0013	EF5 [External Fault (terminal S5)]
0014	EF6 [External Fault (terminal S6)]
0015	EF7 [External Fault (terminal S7)]
0016	EF8 [External Fault (terminal S8)]
0017	FAn [Internal Fan Fault]
0018	oS [Overspeed]
0019	dEv [Speed Deviation]
001A	PGo [PG Disconnect]
001B	PF [Input Phase Loss]
001C	LF [Output Phase Loss]
001D	oH3 [Motor Overheat Alarm (PTC Input)]
001E	oPr [Keypad Connection Fault]
001F	Err [EEPROM Write Error]
0020	oH4 [Motor Overheat Fault (PTC Input)]
0021	CE [MEMOBUS/Modbus Communication Fault]
0022	bUS [Option Communication Error]
0025	CF [Control Fault]
0026	SvE [Zero Servo Fault]
0027	EF0 [Option Card External Fault]
0028	FbL [PID Feedback Loss]
0029	UL3 [Undertorque Detection 1]
002A	UL4 [Undertorque Detection 2]
002B	oL7 [High Slip Braking oL]
0030	Includes oFx Fault [Hardware Fault]
0032	dv1 [Z Pulse Fault]
0033	dv2 [Z Pulse Noise Fault Detection]
0034	dv3 [Inversion Detection]
0035	dv4 [Inversion Prevention Detection]
0036	LF2 [Output Current Imbalance]
0037	STPo [Pull-Out Detection]
0038	PGoH [PG Hardware Fault]
0039	E5 [MECHATROLINK Watchdog Timer Err]
003B	SEr [Too Many Speed Search Restarts]
0041	FbH [Excessive PID Feedback]
0042	EF1 [External Fault (terminal S1)]
0043	EF2 [External Fault (terminal S2)]
0044	oL5 [Mechanical Weakening Detection 1]

Fault code (Hex.)	Name
0045	UL5 [Mechanical Weakening Detection 2]
0046	CoF [Current Offset Fault]
0049	dWFL [DriveWorksEZ Fault]
004A	dWF1 [EEPROM Memory DWEZ Data Error]
004B	dWF2 [DriveWorksEZ Fault 2]
004C	dWF3 [DriveWorksEZ Fault 3]
004E	rF [Braking Resistor Fault]
004F	boL [BrakingTransistor Overload Fault]
0051	LSO [LSO Fault]
0052	nSE [Node Setup Error]
005B	dv7 [Polarity Judge Timeout]
005F	LF3 [Output Phase Loss 3]
0060	UnbC [Current Imbalance]
0061	Uv4 [Gate Drive Board Power Supply Voltage Low]
0083	CPF02 [A/D Conversion Failure]
0084	CPF03 [PWM Motor Failure]
0087	CPF06 [Control Circuit Error]
0088	CPF07 [Terminal Board Connection Error]
0089	CPF08 [EEPROM Serial Communications Error]
008C	CRF11 [RAM Fault]
008D	CRF12 [FLASH Memory Fault]
008E	CPF13 [Watchdog Circuit Exception]
008F	CPF14 [Control Circuit Fault]
0091	CPF16 [Clock Fault]
0092	CPF17 [Timing Fault]
0093	CPF18 [Control Circuit Fault]
0094	CPF19 [Control Circuit Fault]
0095	CPF20 [Hardware Fault (at power ON)]
0096	CPF21 [Hardware Fault (after communication start up)]
0097	CPF22 [A/D Conversion Failure]
0098	CPF23 [PWM Feedback Fault]
0099	CPF24 [Drive Capacity Signal Fault]
009A	CPF25 [Terminal Board not Connected]
009B	CPF26 [ASIC BB Circuit Error]
009C	CPF27 [ASIC PWM Setting Register Error]
009D	CPF28 [ASIC PWM Pattern Error]
009E	CPF29 [ASIC On-delay Error]
009F	CPF30 [ASIC BB ON Error]
00A0	CPF31 [ASIC Code Error]
00A1	CPF32 [ASIC Start-up Error]
00A2	CPF33 [Watchdog Circuit Exception Circuit Fault]
00A3	CPF34 [ASIC Power, Clock Error]

Fault code (Hex.)	Name	Fault code (Hex.)	Name
00A4	CPF35 [External A/D Conversion Failure]	0212	oFb11 [Option Operation Mode Fault (SLMOD)]
00A5	CPF36 [ASIC Reception Error]	0213	oFb12 [CRC Error (Drive receive)]
00A6	CPF37 [ASIC Reception Error]	0214	oFb13 [Frame Error (Drive receive)]
00A7	CPF38 [EEPROM Motor Failure]	0215	oFb14 [Abort Error (Drive receive)]
00A9	CPF40 [Control Circuit Error]	0216	oFb15 [CRC Error (Option receive)]
00AA	CPF41 [Control Circuit Error]	0217	oFb16 [Frame Error (Option receive)]
00AB	CPF42 [Control Circuit Error]	0218	oFb17 [Abort Error (Option receive)]
00AC	CPF43 [Control Circuit Error]	0231	oFb30 [Comm. ID Error]
00AD	CPF44 [Control Circuit Error]	0232	oFb31 [Model Code Error]
00AE	CPF45 [Control Circuit Error]	0233	oFb32 [Checksum Error]
0101	oFA00 [Not supported]	0234	oFb33 [Comm. option timeout waiting for response]
0102	oFA01 [Connection Error]	0235	oFb34 [MEMOBUS/Modbus communications timeout]
0106	oFA05 [A/D Conversion Error]	0236	oFb35 [Drive timeout waiting for response]
0107	oFA06 [Option Response Error]	0237	oFb36 [CI Check Error]
0111	oFA10 [RAM Fault]	0238	oFb37 [Drive timeout waiting for response]
0112	oFA11 [Option Operation Mode Fault (SLMOD)]	0239	oFb38 [Control Command Selection Error]
0113	oFA12 [CRC Error (Drive receive)]	023A	oFb39 [Drive timeout waiting for response]
0114	oFA13 [Frame Error (Drive receive)]	023B	oFb40 [Control Response Selection 1 Error]
0115	oFA14 [Abort Error (Drive receive)]	023C	oFb41 [Drive timeout waiting for response]
0116	oFA15 [CRC Error (Option receive)]	023D	oFb42 [Control Response Selection 2 Error]
0117	oFA16 [Frame Error (option receive)]	023E	oFb43 [Drive timeout waiting for response]
0118	oFA17 [Abort Error (option receive)]	0301	oFC00 [Not supported]
0131	oFA30 [Comm. ID Error]	0302	oFC01 [Connection Error]
0132	oFA31 [Model Code Error]	0303	oFC02 [DuplicateOptions]
0133	oFA32 [Checksum Error]	0306	oFC05 [A/D Conversion Error]
0134	oFA33 [Comm. option timeout waiting for response]	0307	oFC06 [Option Response Error]
0135	oFA34 [MEMOBUS/Modbus communications timeout]	0311	oFC10 [RAM Fault]
0136	oFA35 [Drive timeout waiting for response]	0312	oFC11 [Option Operation Mode Fault (SLMOD)]
0137	oFA36 [CI Check Error]	0313	oFC12 [CRC Error (Drive receive)]
0138	oFA37 [Drive timeout waiting for response]	0314	oFC13 [Frame Error (Drive receive)]
0139	oFA38 [Control Command Selection Error]	0315	oFC14 [Abort Error (Drive receive)]
013A	oFA39 [Drive timeout waiting for response]	0316	oFC15 [CRC Error (Option receive)]
013B	oFA40 [Control Response Selection 1 Error]	0317	oFC16 [Frame Error (option receive)]
013C	oFA41 [Drive timeout waiting for response]	0318	oFC17 [Abort Error (Option receive)]
013D	oFA42 [Control Response Selection 2 Error]	0351	oFC50 [Encoder Option AD Conversion Error]
013E	oFA43 [Drive timeout waiting for response]	0352	oFC51 [Encoder Option Analog Circuit Error]
0201	oFb00 [Not supported]	0353	oFC52 [Encoder Communication Timeout]
0202	oFb01 [Connection Error]	0354	oFC53 [Encoder Communication Data Error]
0203	oFb02 [DuplicateOptions]	0355	oFC54 [Encoder Error]
0206	oFb05 [A/D Conversion Error]	0356	oFC55 [Resolver Error]
0207	oFb06 [Option Response Error]	040F	SCF [Safety Circuit Fault]
0211	oFb10 [RAM Fault]	0413	FAn1 [Drive Cooling Fan Failure]

■ Minor Fault Contents

The following table lists the minor fault codes read using the communications register (007 (Hex.)).

Table 5.10 Minor Fault Contents (007 (Hex.))

Minor fault code (Hex.)	Name	Minor fault code (Hex.)	Name
0001	Uv [DC Bus Undervoltage]	0021	L24v [External 24V Power Supply Depletion]
0002	ov [DC Bus Overvoltage]	0022	oH3 [Motor Overheat (PTC)]
0003	oH [Heatsink Overheat]	0027	FbL [PID Feedback Loss]
0004	oH2 [Drive Overheat Warning]	0028	FbH [Excessive PID Feedback]
0005	oL3 [Overtorque 1]	002A	dnE [Drive Disabled]
0006	oL4 [Overtorque 2]	002B	PGoH [PG Hardware Fault]
0007	EF [FWD/REV Run Command Input Error]	0031	E5 [MECHATROLINK Watchdog Timer Err]
0008	bb [Baseblock]	0032	AEr [Station Address Setting Error (CC-Link, CANopen, MECHATROLINK)]
0009	EF3 [External Fault (terminal S3)]	0033	CyC [MECHATROLINK CommCycleSettingErr]
000A	EF4 [External Fault (terminal S4)]	0034	HCA [Current Alarm]
000B	EF5 [External Fault (terminal S5)]	0035	LT-1 [Cooling Fan Maintenance Time]
000C	EF6 [External Fault (terminal S6)]	0036	LT-2 [Capacitor Maintenance Time]
000D	EF7 [External Fault (terminal S7)]	0039	EF1 [External Fault (terminal S1)]
000E	EF8 [External Fault (terminal S8)]	003A	EF2 [External Fault (terminal S2)]
000F	FAn [Internal Fan Fault]	003B	SToF [Safe Disable Signal Input]
0010	oS [Overspeed]	003C	STo [Safe Disable Signal Input]
0011	dEv [Speed Deviation]	003D	oL5 [Mechanical Weakening Detection 1]
0012	PGo [PG Disconnect]	003E	UL5 [Mechanical Weakening Detection 2]
0014	CE [MEMOBUS/Modbus Communication Err]	0042	TrPC [IGBT Maintenance Time (90%)]
0015	bUS [Option Communication Error]	0043	LT-3 [SoftChargeBypassRelay MainteTime]
0016	CALL [Serial Comm Transmission Error]	0044	LT-4 [IGBT Maintenance Time (50%)]
0017	oL1 [Motor Overload]	0045	boL [BrakingTransistor Overload Fault]
0018	oL2 [Drive Overloaded]	0049	dWAL [DriveWorksEZ Fault]
001A	EF0 [Option Card External Fault]	004A	dWA2 [DriveWorksEZ Alarm 2]
001B	rUn [Motor Switch during Run]	004B	dWA3 [DriveWorksEZ Alarm 3]
001D	CALL [Serial Comm Transmission Error]	0081	EP24v [External Power 24V Supply]
001E	UL3 [Undertorque Detection 1]	0082	LoG [Log Communication Fault]
001F	UL4 [Undertorque Detection 2]	0085	bAT [Keypad Battery Voltage Low]
0020	SE [MEMOBUS/Modbus Comm Test Mode Err]		

◆ Error Code

■ MEMOBUS/Modbus Communications Error Code List

The following table lists the MEMOBUS/Modbus communications error codes.

When an error occurs, resolve the cause of the error and restart the communications.

Table 5.11 MEMOBUS/Modbus Communications Error Code

Error Code (Hex.)	Name	Cause
01	Function code error	A function code other than 03, 08, and 10 (Hex.) was set from PLC.
02	Register Number Error	- No register No. is registered while attempting to access. - Starting numbers other than 0001 or 0002 (Hex.) were set when broadcasting.
03	Bit Count Error	- Read and write data quantities have exceeded the 1 to 16 range. (Command message data quantity is disabled.) - In the write mode, the number of bytes in the message is not the number of data $\times$ 2.
21	Data Setting Error	- Writing control data or parameters made settings go outside the allowable setting range. - A parameter setting error occurred when writing a parameter.
22	Write Mode Error	- Attempted to write a disabled parameter during run. - When <i>CPF06 [EEPROM Data Error]</i> occurs, a parameter other than the following was attempted to be written from the master. – <i>A1-00 [Language Selection]</i> – <i>A1-01 [Access Level Selection]</i> – <i>A1-02 [Control Method Selection]</i> – <i>A1-03 [Initialize Parameters]</i> – <i>A1-04: [Password]</i> – <i>A1-05: [Password Setting]</i> – <i>E1-03 [V/f Pattern Selection]</i> – <i>o2-04 [Drive Model Selection]</i> - Writes the read-only data.
23	DC Bus Undervoltage Write Error	During <i>Uv [DC Bus Undervoltage]</i> , a <i>Uv</i> write disabled parameter was written.
24	Error Writing Data during parameter processing	Attempted to write a parameter from the master during parameter processing on the drive side.
25	Writing into EEPROM disabled	EEPROM write was executed from the MEMOBUS/Modbus communications while writing into EEPROM write is disabled. When this error occurs, a message is displayed and operation continues.

■ Slave Non Reply

In the following cases, the slave ignores the command message from the master and sends no response message.

- When a communications error (overrun, framing, parity, CRC-16) is detected in the command message.
- When the slave address inside the command message and the slave address for the drive side do not match (the slave address of the drive is set in *H5-01 [Drive Node Address]*).
- When the time interval between the data of which the message is composed exceeds the 24 bit length
- When the data length for the command message is inaccurate

Note:

- When *CALL [Serial Comm Transmission Error]* appears on the keypad, refer to “Troubleshooting” to resolve the cause of the abnormality, and try to execute communications again. If *CALL* does not appear on the keypad, check for the presence of an error and the type of error in *U1-19 [MEMOBUS/Modbus Error Code]*.
- When write function code is executed and when the slave address that is set inside the command message is 00 (Hex.), write is executed by all slaves, but response messages are not sent to the master.

